Timeout

An NBA play recommender

Pause a basketball game at any moment. Timeout looks at where all ten players and the ball are, and draws on the frame the best thing the offense could do next — an arrow, a highlighted player, and a number telling you how many points that choice is worth. Then it explains why, in a sentence.

one State schema — identical from broadcast CV or SportVU tracking

A complete guide — read this and you understand the whole project.

1. What Timeout is

Timeout is a tool that watches basketball footage and acts like an assistant coach who never blinks. Whenever you pause, it answers one question: **of everything the player with the ball could do right now — shoot, drive, pass to a teammate, set a screen — which is worth the most?**

The important idea is what it is *not*. It is not trying to predict what *will* happen. It is grading the *options*. Think of it as a second opinion with a number attached: not “he is going to pass left,” but “a pass to the corner is worth about 1.1 points; a contested shot here is worth about 0.8, so the pass is the better play.”

Everything it shows is measured, not guessed. Player positions come from the video. The scores come from models trained on thousands of real NBA possessions. The written explanation is only allowed to repeat numbers the model already produced — it can’t make things up. And when the picture is too unclear to be sure (a replay, a close-up, players hidden behind each other), it says so and shows nothing rather than guess.

The one formula, in plain words

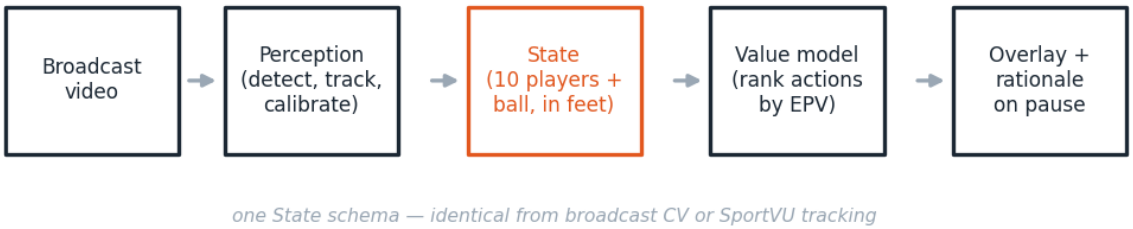
Every option is scored the same way. The value of an action is: how likely it is to succeed, times how good the resulting situation is — plus, for the times it fails, how bad the turnover is. Written out:

```
value(action) = P(success) × value(good outcome) + (1 − P(success)) × value(turnover)
```

“Value” here is measured in **expected points** — the average number of points that situation is worth. This single number lets you compare a shot, a pass, and a drive on the same scale, which is the whole point.

2. How it works, end to end

The system is a pipeline: raw video goes in one end, a drawn-on recommendation comes out the other. There are four stages.



The pipeline. The middle box — the State — is the heart of the design.

Stage 1 — Perception. Look at the video and figure out where everyone is. Detect the players and the ball, follow them frame to frame, and convert their positions from “pixels on screen” into “feet on the court.”

Stage 2 — The State. Package all of that into one tidy object: ten players and the ball, in court coordinates, with their speeds, who is guarding whom, how much space there is, and a confidence score. This object is the contract the rest of the system depends on.

Stage 3 — The value model. Given the State, list every sensible action the ball-handler could take and score each one in expected points.

Stage 4 — The app. Draw the best action on the paused frame, show the ranked list of alternatives, and — on request — explain the choice in plain English or answer questions in a chat box.

Why the State matters so much

The clever part of the design is that the State looks *exactly the same* whether it came from broadcast video or from the NBA’s official player-tracking data (called SportVU). Because both sources produce an identical State, the scoring models — which were trained on the clean official data — work without any changes on messy video. The hard, unreliable part (reading video) is walled off from the smart part (scoring), and they meet at one clean hand-off.

3. Seeing the game (perception)

Turning a video into court positions is the messiest job in the project. Here is each piece, in order.

Calibration — teaching it where the court is

A camera can be anywhere, at any angle. To convert screen pixels into real court feet, the system needs to know how the court is positioned in the shot. You do this once per camera angle: a small court diagram highlights a landmark (say, the corner of the paint), and you click the matching spot in the video. After 5–8 clicks it works out the full mapping (a *homography*) and draws the court lines back onto the frame so you can check the fit. As the camera pans, the mapping is carried along automatically by tracking the movement between frames.

Calibrated shot	Landmarks clicked	Accuracy (reprojection error)
early_a.json	8	0.842 ft
early_b.json	5	0.144 ft
calib.json	7	1.325 ft

Reprojection error is how far off the mapping is when checked against the clicked points — under about 1.5 ft is good; these are all well within that.

Per-frame calibration — the trained keypoint model

A click calibration is solved on *one* frame and carried across the shot by optical flow, so its accuracy decays with distance from that frame. Measured on a real shot, the carried mapping drifts from 0.18 ft to **16.8 ft** over eleven seconds — catastrophically wrong about seven seconds after its anchor. The fix is a model that finds the court landmarks independently in every frame.

The obstacle is labels: hand-marking broadcast frames is exactly the work the click calibration was invented to avoid. So the labels are generated from the calibrations that already exist. A solved calibration *is* a pixel-to-court correspondence, so inverting it places all 26 canonical landmarks on the frame — including the ones nobody clicked. Two harvesting strategies follow from that:

- **Flow propagation** walks outward from the clicked frame and labels every frame it passes. It gives real camera motion, but inherits the tracker's drift — so each frame is gated on the tracker's own reprojection error and dropped once it degrades.
- **Homography jitter** warps the verified anchor frame by a random perspective transform and pushes the labels through the same matrix. The labels stay *exact* however aggressive the warp, because the warp *is* the label transform. This manufactures new camera geometry without manufacturing label error.

The model is a small (1.5M-parameter) encoder-decoder that outputs a heatmap per landmark. Heatmaps rather than direct coordinates for one practical reason: the height of each peak is a natural confidence, which is exactly what the homography solver already expects, so an occluded landmark arrives with a low score and is dropped by the same gate the rest of the system uses.

A landmark that is off-frame is supervised toward an *empty* heatmap rather than being excluded from training. Visibility is known geometrically, so absence is a fact worth teaching; leaving it unsupervised lets the network emit a confident peak for a landmark that is nowhere in the picture.

The trap: self-consistency is not correctness

The most useful lesson from building this is a negative one. Reprojection error only measures whether a homography agrees with the correspondences it was fitted to. A model that hallucinates a *mutually consistent* set of landmarks therefore reports a healthy error while projecting a court nowhere near the real one — which is what happened, and was only caught by drawing the result on the frame and looking at it. Worse, four points fit a homography with zero residual by construction, so a model that finds four landmarks and hallucinates the rest scores *better* than one that finds twenty.

Acceptance therefore requires an independent, physical check as well: a foot of court spans 17–56 px on the measured click calibrations, versus 1.5–5 px on hallucinated ones — the camera has effectively been placed in the next building. Scale is sampled only where the court is actually visible, because on a zoomed shot the far baseline sits beyond the vanishing point where scale legitimately explodes. The check accepts all seven real calibrations and rejects the bad predictions.

Where it stands

Trained on 683 auto-labelled frames from six usable calibrated shots across three games, validated on a whole held-out camera shot. Within a shot it has seen, the model holds 1.55–1.74 ft while the carried calibration collapses to 16.8 ft. On a camera angle it has never seen, none of its predictions survive the acceptance gate. Training loss keeps falling while held-out error plateaus, which is overfitting to six camera angles: **the bottleneck is labelled shots, not epochs or architecture**. Every additional calibrated shot feeds the same harvesting pipeline unchanged. The feature is therefore opt-in and off by default, and the gate refuses its held-out output rather than letting it corrupt the overlay.

Two calibrations were excluded from training: each had only four clicked points, which constrains nothing, and one of them turned out to be a close-up of a player rather than a view of the court.

Detection — finding players and the ball

A detector scans each sampled frame and boxes the players and the ball. The big lesson learned here: a generic “find people” detector is the wrong tool, because it also boxes the crowd, the bench, coaches, and referees — sometimes 30–40 “people” when only 10 are playing. The project now uses a purpose-built basketball detector (a hosted Roboflow workflow) with a dedicated *player* class that ignores the crowd, a separate *referee* class so officials can be dropped, and reliable *ball* detection. On the test clip this took the count from a noisy 30+ down to a clean 10 on-court players.

Following, teams, and names

- **Tracking:** each player keeps the same identity from frame to frame, so paths are smooth and the ball-handler doesn’t flicker between people.

- **Teams:** jersey colours are clustered into two groups, so the system knows which five are on offense and which five are defending.
- **Jersey numbers:** an optical-character-reader reads the numbers it can see and votes across the whole shot, so a player becomes “#7” and the explanations can name him. Players facing away stay unnamed rather than being guessed.
- **Ball tracking over time:** the ball is small and blurs, so it is missed on some frames. Instead of only trusting a single frame, the system models the ball’s motion and coasts its position through the gaps, so every pause has a ball — and therefore a known handler.

Which detector, and why it matters

Detectors are interchangeable: each one’s own class *names* are mapped onto the project’s schema, so swapping detectors needs no code change.

- **Hosted basketball workflow (recommended).** A serverless player-detection workflow whose *player* class finds the ten on-court players and not the crowd, with a separate *referee* class that can be dropped and reliable *ball* detection. One network call per sampled frame, so a larger frame stride and several concurrent workers are used; transient timeouts are retried, and a single unreachable frame is skipped rather than failing the whole build. The API key is read from the environment and never stored.
- **Local YOLO (default, offline).** Generic COCO weights, or a fine-tuned basketball model if one is present. Runs on the GPU when a matching CUDA build is installed.

Robustness on messy broadcast pixels

Several steps stand between a raw frame and a clean ten-player state with the handler on the right person.

- **Referee exclusion** — the detector’s referee class is dropped, with a conservative grey-uniform appearance check as a backup, deliberately tuned to risk missing a referee rather than removing a real player.
- **Roster gate** — a track survives only if its *median* court position is that of an interior on-floor player, which removes anything projected onto the sidelines.
- **Temporal ball tracking** — a constant-velocity model coasts the ball through the frames where the detector misses it.
- **Handler voting** — possession is mode-filtered over neighbouring frames, so the ball-handler cannot flicker between players.
- **Camera-cut truncation** — a shot’s mapping is valid only until the next cut, so processing stops there; a multi-shot build anchors a fresh calibration in each shot and merges the results into one timeline.

Coverage — more calibrated shots, more of the game

One calibration covers one camera shot, from its click-time to the next cut. Calibrating a frame in each shot and passing them together lets the build process each shot bounded by the next and merge them into a single timeline. More calibrations is simply more coverage — and, as section 3 explains, also the one input that would make the keypoint model generalise.

The confidence gate — knowing when to stay quiet

Every State carries a confidence score built from how many players were tracked, how good the court mapping is, and whether the ball was seen. Below a threshold the app withholds the recommendation and tells you why (replay, close-up, occlusion, or a stretch of video that wasn’t calibrated). Showing nothing is treated as better than showing a wrong arrow.

4. Scoring the options (the value model)

Once there is a clean State, the system lists the legal actions for the ball-handler (up to about 13 of them: shoot, drive left/right, pass to each teammate, set or use a screen, reset) and scores each with the formula from section 1. Three ingredients feed that formula.

The three sub-models

- **Shot model** — how likely a shot is to go in, given who is shooting, from where, and how closely guarded. It blends a per-player shooting history with the situation.
- **Pass model** — how likely a pass is to reach its target without being stolen.
- **Drive model** — how likely a drive to the basket gets there successfully.

These three are gradient-boosted models (LightGBM) — fast, reliable classifiers trained on real NBA outcomes.

The situation value, V(s)

The formula also needs the value of the *resulting* situation — e.g. how good it is to have the ball in the corner with a step of space. That is a neural network called V(s). It is built to be *order-independent*: it treats the players as a set, so shuffling their order can't change the answer. It runs on a GPU when one is available.

Two honesty rules

- **Actions are objects, never coordinates.** The model outputs "pass to the player in the left corner," and only the renderer turns that into an arrow — and the arrow's endpoints are always real tracked positions, never invented ones.
- **Never invent a missing defender.** If a defender can't be seen, the system lowers its confidence instead of imagining one, and may withhold the recommendation.

5. The web app

The product is a local web page with two parts.

The studio (setup wizard)

You upload a clip from your computer or paste a YouTube link. It grabs the footage, opens a calibration step where you click the court landmarks (and can click where the ball is in the first frame to seed tracking), and then builds the recommendations. You can calibrate several camera angles to cover more of the game.

The player

- Play the footage and **pause anywhere**; the best action is drawn on the frame with an arrow, a highlighted player, and its expected-points number.
- A ranked list of **alternative actions** sits beside the video — click any to draw it instead.
- Ask **why** and read a one-paragraph rationale for the chosen action.
- A **chat assistant** answers free-form questions ("why not shoot?", "show the drive", "pass to #7") and can change the drawn play. It runs through a language model (Groq or Claude) when a key is set, and falls back to built-in rules otherwise.

Crucially, the app does **no live inference**. All the heavy work is done once, up front, and saved to a file the page reads. So pausing is instant.

Because each drawn overlay is computed for its own frame, the app refuses to reuse one recommendation on a far-away moment — that is what previously made an arrow appear frozen and point into the stands during an uncalibrated opening stretch. It now shows an honest "outside the calibrated segment" note there instead.

6. How the models were trained

The scoring models learn from two kinds of data.

- **Synthetic data** — a built-in generator makes fake-but-realistic possessions with a known "right answer," so the whole pipeline can be tested and measured with no downloads and no GPU.
- **Real data** — NBA 2015-16 SportVU player-tracking joined to real shot and play-by-play outcomes. The models here were trained on roughly **240 games** (about 25,000 shots, 45,000 passes, 49,000 drives, and 600,000+ situation snapshots), split so that test games are never seen during training.

The possession parser recovers realistic NBA rates on its own — about 100 possessions per team per game at roughly one point per possession — which is a good sign the inputs are sane before any model is trained.

7. Benchmark results

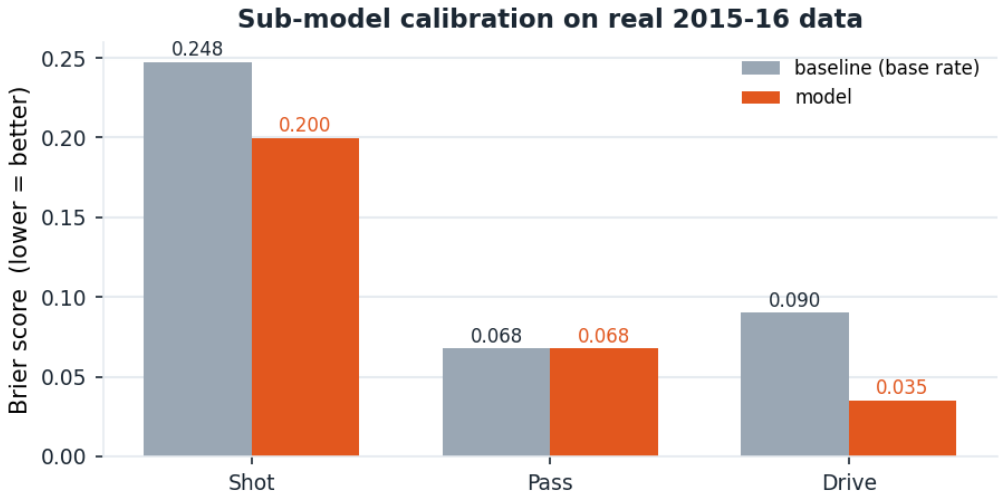
Every number below was produced by the project's own scripts (`scripts/benchmark.py` and the training runs).

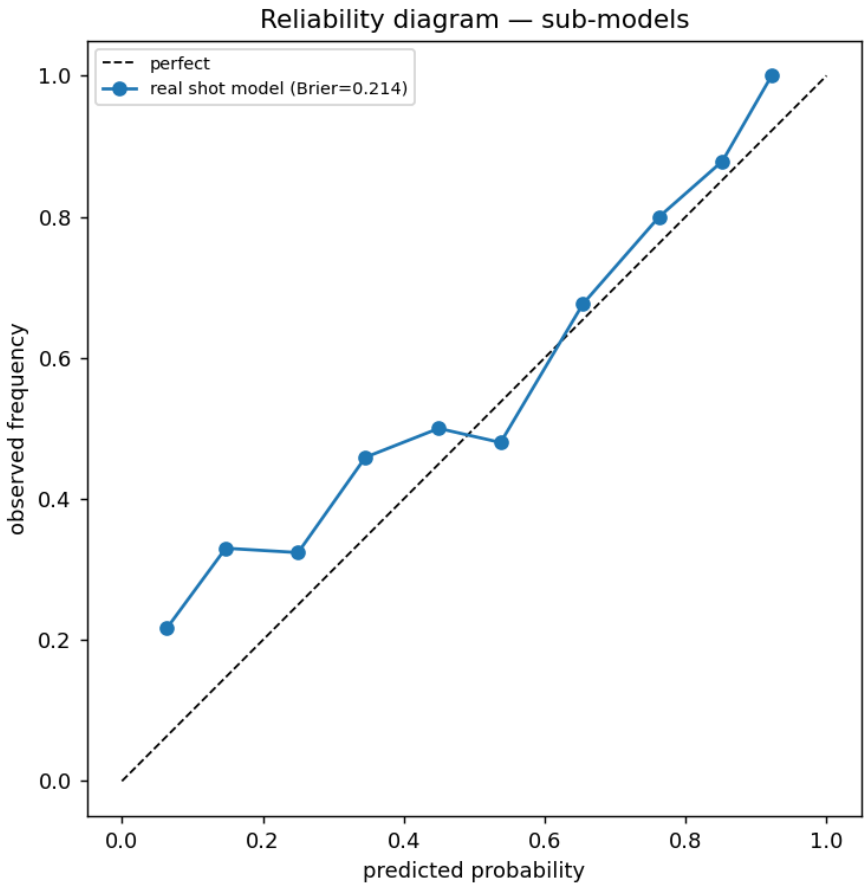
Does the scoring actually work? (real 2015-16 data)

Sub-model	Score (Brier)	Baseline	Verdict
Shot	0.200	0.248	beats baseline
Pass	0.068	0.068	ties baseline
Drive	0.035	0.090	beats baseline

Brier score measures how well-calibrated a probability is — lower is better, and the baseline is "just guess the average rate every time." The shot and drive models clearly beat that; pass completion ties it, because passes succeed ~90%

of the time and the flat guess is already hard to beat.





Shot-model reliability: predicted make-probability vs. what actually happened. Closer to the diagonal is better.

Does it pick good actions? (simulated, vs. naive strategies)

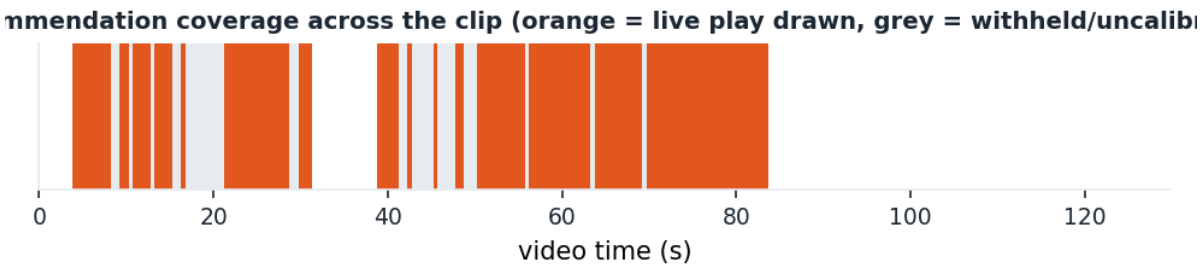
Strategy	Right or defensible	Clearly wrong	Avg. points lost vs. best

Timeout	86%	4%	0.058
Always pass to most open	40%	48%	0.274
Always shoot	30%	55%	0.337
Random	20%	55%	0.352

Against a known ground-truth generator, Timeout's top pick is right or defensible about 86% of the time and clearly wrong only 4%, losing far fewer expected points than any simple rule.

Does it work on real footage? (the test clip)

What was measured	Result
Clip	[1280, 720] at 30.0 fps, 129.8 s
Pause points with a recommendation	111 of 146 (76.0%)
On-court players detected per pause	median 10 (range 7–13)
Pauses with a live ball detection	90.1%
Pauses with a resolved ball-handler	100.0%
Pauses with the rim located	100.0%



A median of exactly 10 players per pause is the headline: it means the detector is seeing the five-on-five action and not the crowd.

Is it fast enough?

Scoring one paused moment — listing all 13 candidate actions and running every model — takes about **37.83 ms** (median; 41.51 ms at the 90th percentile). The design budget was 2 seconds, so there is enormous headroom, and because results are pre-computed the app itself pauses instantly.

Does per-frame calibration beat the carried one?

Measured over eleven seconds of a real camera shot, comparing the click calibration carried forward by optical flow against a homography solved independently on every frame by the keypoint model:

	carried calibration	per-frame model
Error at the start of the window	0.18 ft	1.55 ft

Error eleven seconds later	16.8 ft	1.74 ft
Frames passing the acceptance gate	—	30% in-domain, 0% held-out

The carried mapping is stable at first and then collapses; the model does not drift at all. That is the whole case for the approach. The last row is the honest counterweight — on a camera angle it has never seen, none of the model's output survives the geometric check, so the feature stays off by default until more shots are calibrated.

And the whole system is covered by an automated test suite: **83 tests pass** end to end.

8. Honest limitations

A description isn't complete without the rough edges. These are known and documented, not hidden.

- **The situation model V(s) is weak on real data.** The shot/pass/drive models are solid, but predicting the exact value of a single state from single-sample outcomes is hard; V(s) barely beats a flat average. It needs smoothed multi-step returns and more data to improve. The action *ranking* still works because it leans mostly on the three strong sub-models.
- **Calibration is still manual in practice.** You click court landmarks once per camera angle. An automatic court-line detector was tried and set aside — broadcast courts mix light and dark lines with logo and jersey interference. The trained keypoint model (section 3) is built, tested and wired in, and it demonstrably removes the drift the carried calibration suffers; but on six calibrated shots it does not yet generalise to an unseen camera angle, so it stays off by default. The bottleneck is labelled shots, and calibrating more of them needs no code change.
- **Coverage has gaps.** Recommendations exist only inside calibrated camera shots; between them (and after a cut) the app honestly shows nothing rather than a stale arrow. More calibrations = more coverage.
- **The action mix can skew.** On the test clip the top pick is "set a screen" most of the time; whether that is genuinely optimal or a model bias is the next thing to investigate — a scoring question, not a detection one.

9. Deployment

The app is deployed as a public web page, and the pre-compute pass can run as a batch job on AWS. The two are independent: the page is meant to stay up and costs cents a month, while the job runs once per clip and leaves nothing running afterwards.

The web app — object storage plus a CDN

The page is three files: the HTML, the pre-computed recommendations, and the clip itself. In the intended design they sit in a *private* bucket that only the CDN can read, signed through an origin access control, with the CDN serving byte-range requests so the video can be scrubbed. Uploads carry explicit content types and cache headers — the video is immutable and cached for a year, the recommendations are not cached at all — and each deploy invalidates the edge cache so a rebuild is

visible immediately.

There is a deliberate second path. A brand-new cloud account cannot create CDN distributions until the provider verifies it, which fails the stack outright. Rather than block the demo on a support queue, the deploy script takes a flag that swaps in a plain static-website bucket: live in about a minute, at the cost of being publicly readable and served over plain HTTP. Both templates share the same bucket name and logical resource, so once the account is verified the same command without the flag upgrades the stack in place — the bucket goes private, the CDN appears in front of it, and nothing is torn down.

The chat assistant degrades on purpose here: a static origin has no endpoint to answer it, the page's request fails, and it falls back to the built-in rule-based assistant. That is better than shipping a hosted API key.

The pre-compute pass as a batch job

Batch processing job, not a hosted inference endpoint — the pipeline runs once per clip and writes a JSON file; there is no per-request inference anywhere in the product, so a real-time endpoint would idle at GPU prices for a workload that never makes an online call. The image installs CPU-only tensors on purpose: the GPU wheels add gigabytes, and on a home connection the upload dominates the wall-clock cost of the whole exercise.

The platform is directory-driven rather than argument-driven: declared inputs are downloaded into the container before it starts, and whatever lands in the declared output directory is uploaded when it exits. A small entrypoint adapts those paths onto the existing command-line interface, so the same pipeline runs unchanged locally and in the cloud. The execution role is scoped to this project's bucket and repository rather than using the broad managed policy, and a maximum-runtime setting acts as a spend guard.

Building the image without a local Docker daemon

Building the job image normally needs Docker running locally. On Windows Home that means the WSL2 backend, because the Hyper-V backend is not offered on that edition — and WSL2 can fail in ways no Docker-side fix reaches. On the development machine the host compute service refuses to create *any* virtual machine, so every Linux distribution fails identically and reinstalling Docker changes nothing. The container path is unavailable locally while the pipeline it builds remains perfectly runnable in the cloud.

The answer is to move the build rather than fix the laptop: a managed build project compiles the image in the cloud and pushes it to the registry. The Dockerfile is unchanged and the resulting tag is identical — only the machine running the build moves. It is also faster, because the layers reach the registry from inside the cloud rather than over a home uplink. The build context is packed from exactly the paths the Dockerfile copies, since shipping the repository would mean uploading tens of gigabytes of clips.

A recurring theme for a new cloud account: services are gated one at a time. The CDN needs account verification; the build service starts with a concurrency quota of zero, so the first build fails with a limit error before the project is ever exercised. Neither is a misconfiguration, and both are worth checking before debugging one's own templates.

10. Running it yourself

Install and run the test suite (no data or GPU needed):

```
pip install -e .
python -m pytest -q
```

Build and watch the web app on a clip:

```
# calibrate one frame per camera angle (a court diagram guides your clicks)
python scripts/calibrate.py --video clip.mp4 --time 39 --out calib.json
# pre-compute recommendations, then serve the player
python scripts/build_webapp.py --video clip.mp4 --shots calib.json
python scripts/serve_webapp.py # open the printed URL
```

For the best detection, set a Roboflow key (read from the environment, never stored) and add `--detector roboflow`. For the chat assistant, set a Groq or Anthropic key. Re-run the whole benchmark with `python scripts/benchmark.py`.

Train the court-keypoint model from calibrations you have already clicked — no new labelling — then check it against the carried calibration and, if you want it, build with per-frame homography:

```
python scripts/train_keypoints.py \
    --pair clip.mp4 shot1.json shot2.json shot3.json
python scripts/eval_keypoints.py --video clip.mp4 --calib shot1.json --render
out/kp.png
python scripts/build_webapp.py --video clip.mp4 --shots shot1.json --keypoint-model
```

The evaluation script writes an overlay image with both mappings drawn on the same frames. Look at it — as section 3 explains, the numeric error alone cannot tell you whether the court is in the right place.

Deploy the built page, and run the pre-compute pass in the cloud:

```
# static hosting; drop -NoCloudFront once the account is CDN-verified
./deploy/aws/webapp/deploy.ps1 -Source out/webapp -NoCloudFront
# build the job image without a local Docker daemon, then run the job
python deploy/aws/sagemaker/build_image_codebuild.py
python deploy/aws/sagemaker/run_job.py --video clip.mp4 --calib shot1.json
```

11. Glossary

Term	Plain meaning
Expected points (EPV)	The average points a situation or action is worth — the common yardstick that lets a shot, pass, and d
State	The tidy object describing all ten players and the ball in court feet, plus speeds, matchups, spacing, ar
Homography	The mathematical mapping that converts a point on the screen into a point on the real court, worked o
Sub-model	One of the three trained classifiers — shot, pass, or drive — that estimates the chance an action succ
V(s)	The neural network that estimates how valuable a resulting situation is.
Brier score	A measure of how well-calibrated a predicted probability is; lower is better.
Confidence gate	The rule that withholds a recommendation when the picture is too unclear to trust.
SportVU	The NBA’s official optical player-tracking data, used to train the models.

Reprojection error	How far a mapping misses the points it was fitted to, in feet. A measure of self-consistency — a low va
Keypoint heatmap	The model’s output per court landmark: a picture of where it believes that landmark is, whose peak he
Held-out shot	A whole camera angle kept out of training, so the score measures generalisation to an uncalibrated sh

Timeout — half-court offense, ball-handler decisions, one honest recommendation per pause.